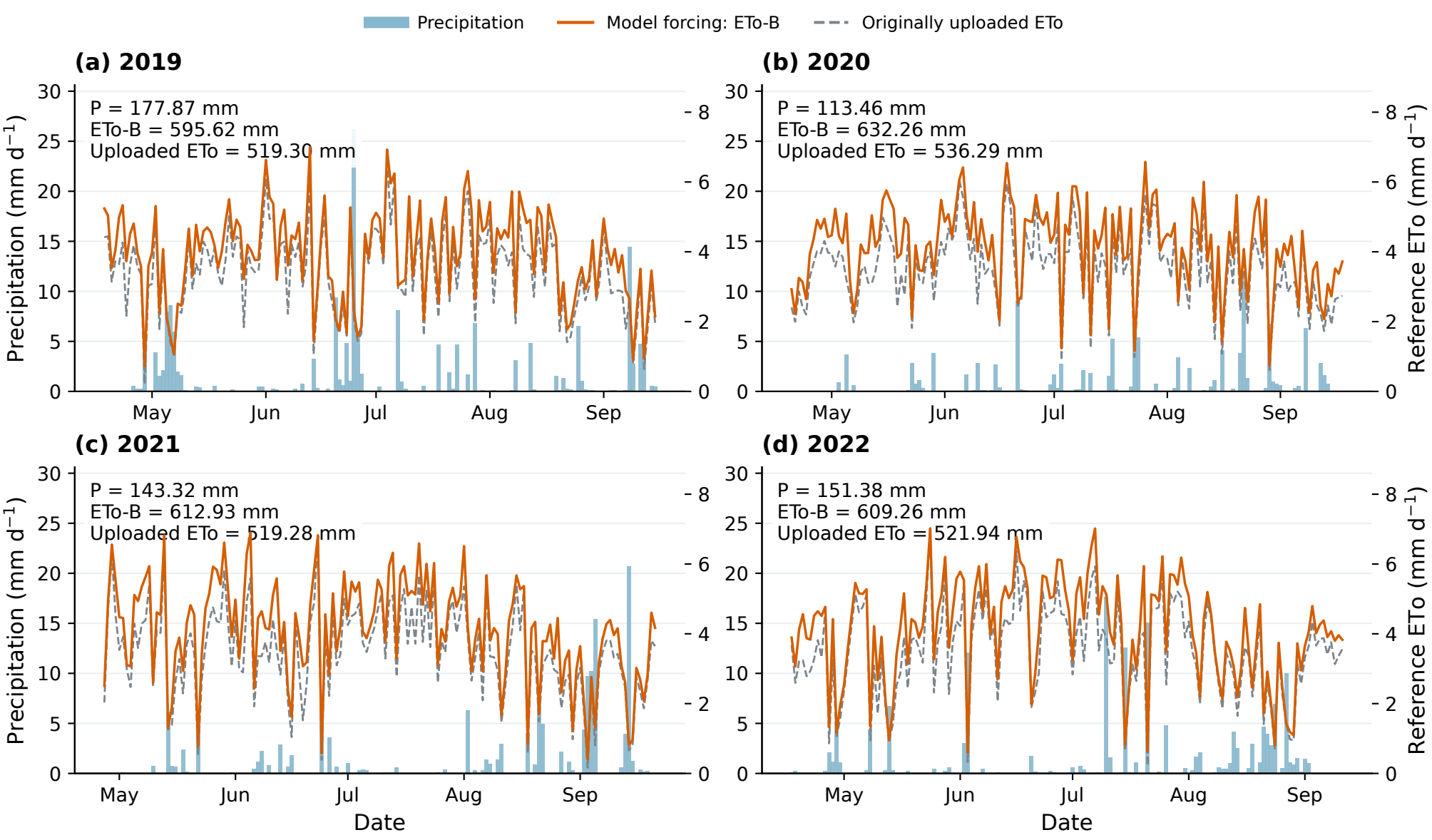
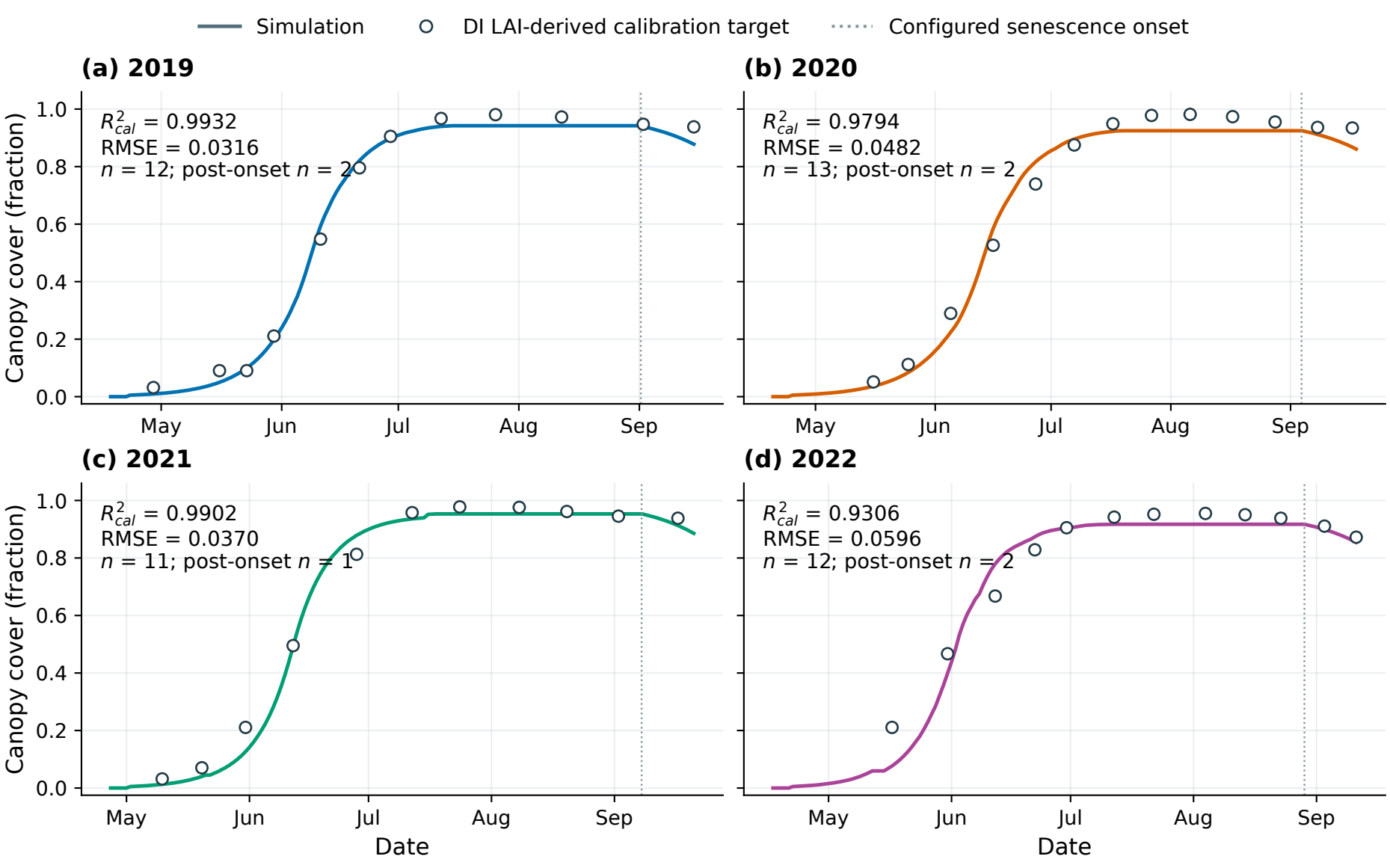
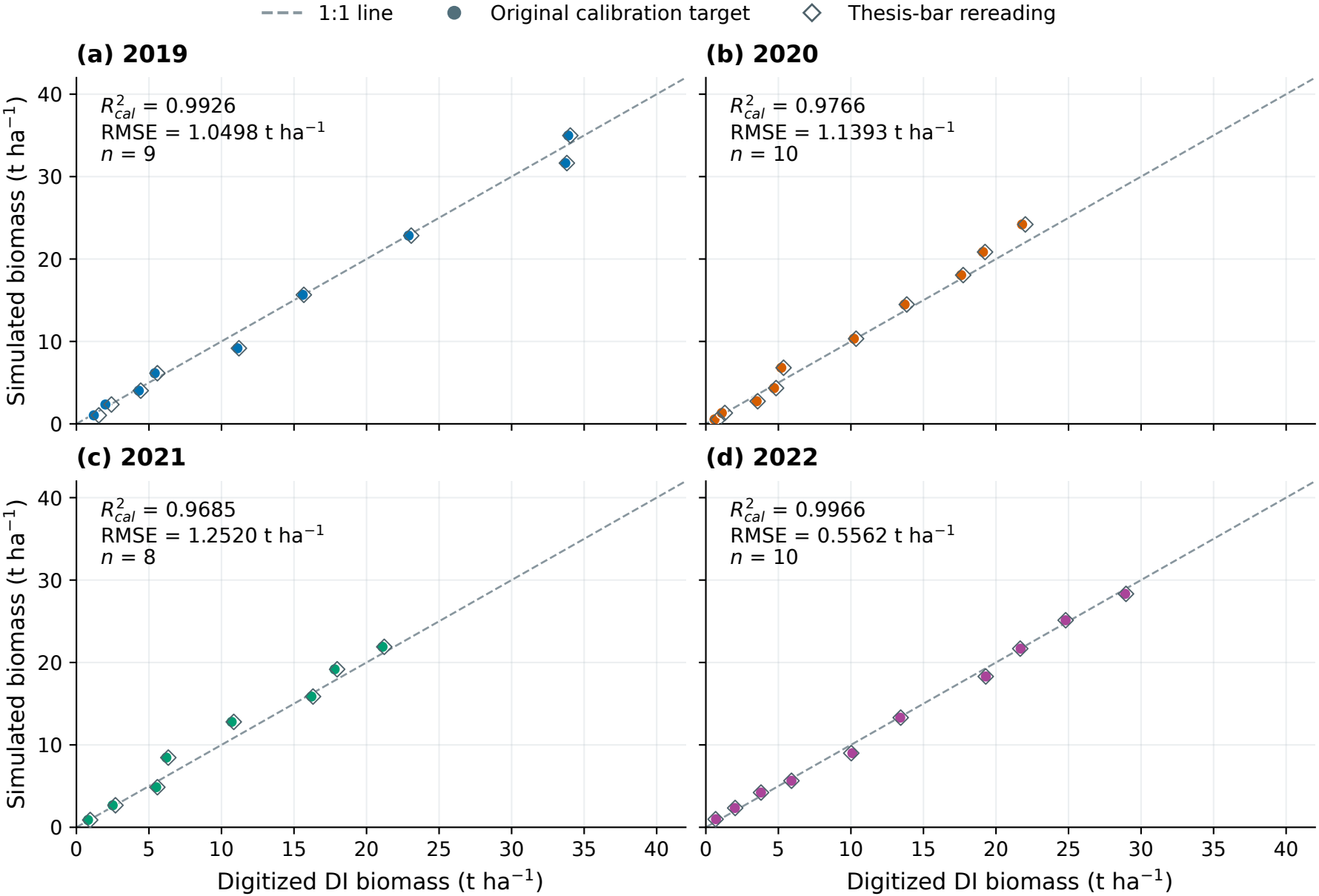
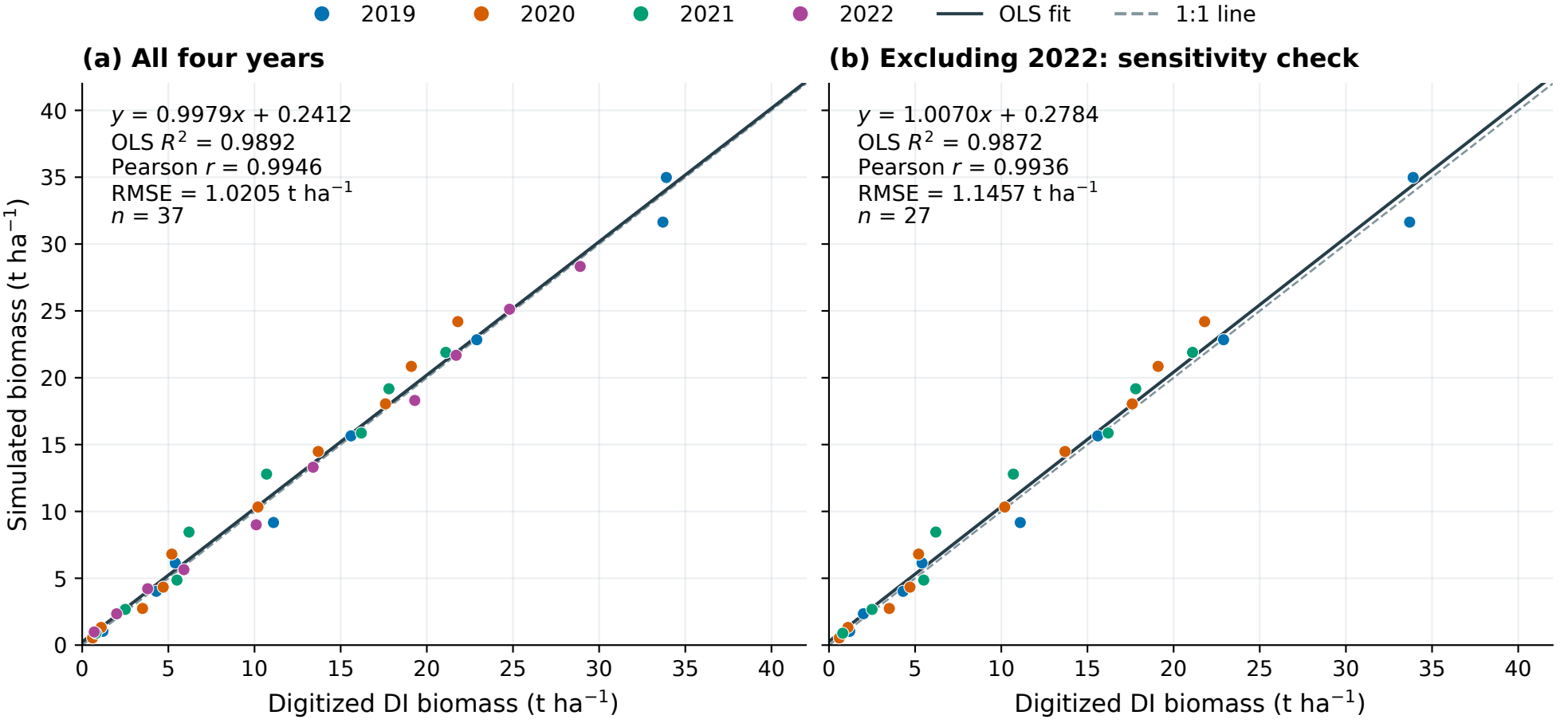
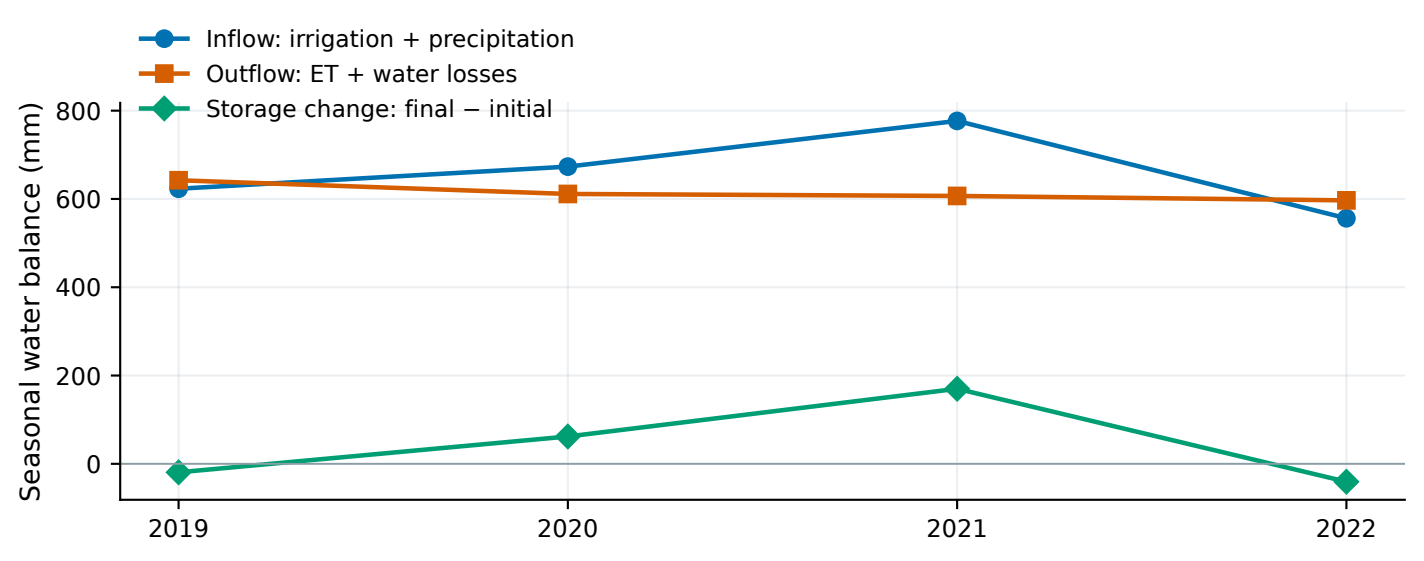
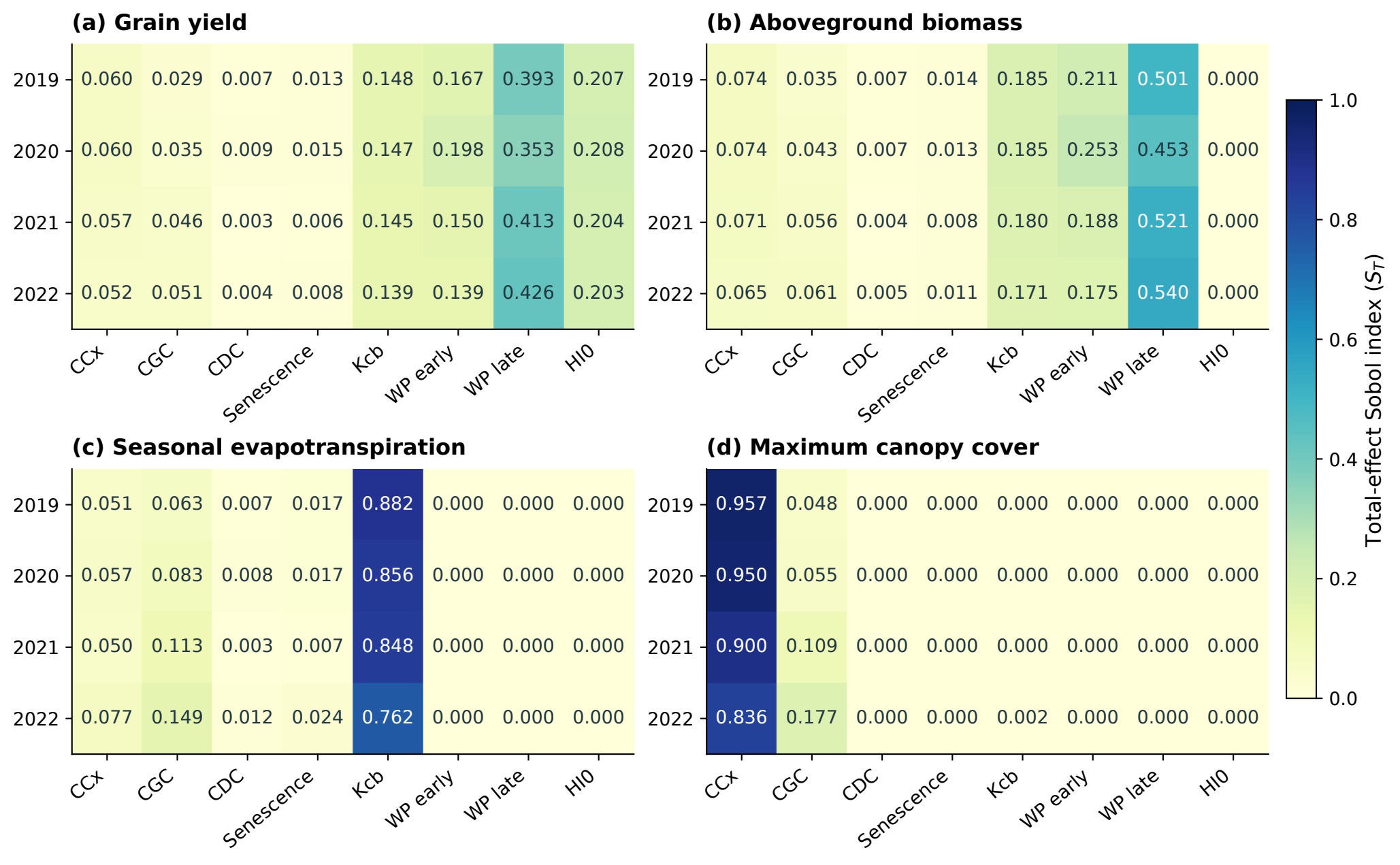

Precipitation
Model forcing: ETo-B
Originally uploaded ETo
(a) 2019
P = 177.87 mm
ETo-B = 595.62 mm
Uploaded ETo = 519.30 mm
(b) 2020
P = 113.46 mm
ETo-B = 632.26 mm
Uploaded ETo = 536.29 mm
(c) 2021
P = 143.32 mm
ETo-B = 612.93 mm
Uploaded ETo = 519.28 mm
(d) 2022
P = 151.38 mm
ETo-B = 609.26 mm
Uploaded ETo = 521.94 mm
Precipitation (mm d^{-1})
Reference ETo (mm d^{-1})
May
Jun
Jul
Aug
Sep
Date

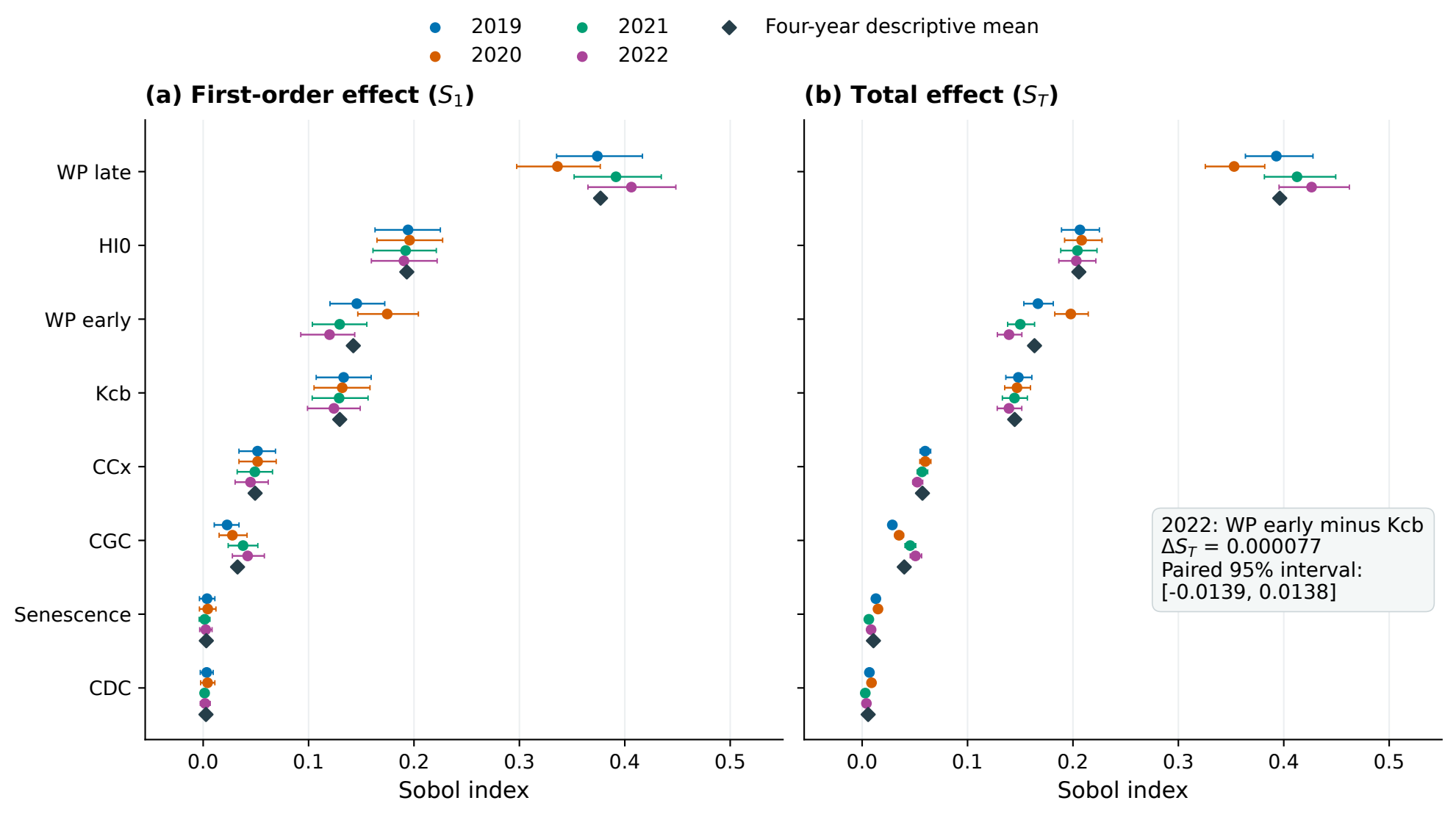

Simulation
DI LAI-derived calibration target
Configured senescence onset
(a) 2019
R^2_{cal} = 0.9932
RMSE = 0.0316
n = 12; post-onset n = 2
(b) 2020
R^2_{cal} = 0.9794
RMSE = 0.0482
n = 13; post-onset n = 2
(c) 2021
R^2_{cal} = 0.9902
RMSE = 0.0370
n = 11; post-onset n = 1
(d) 2022
R^2_{cal} = 0.9306
RMSE = 0.0596
n = 12; post-onset n = 2
Canopy cover (fraction)
0.0
0.2
0.4
0.6
0.8
1.0
May
Jun
Jul
Aug
Sep
Date

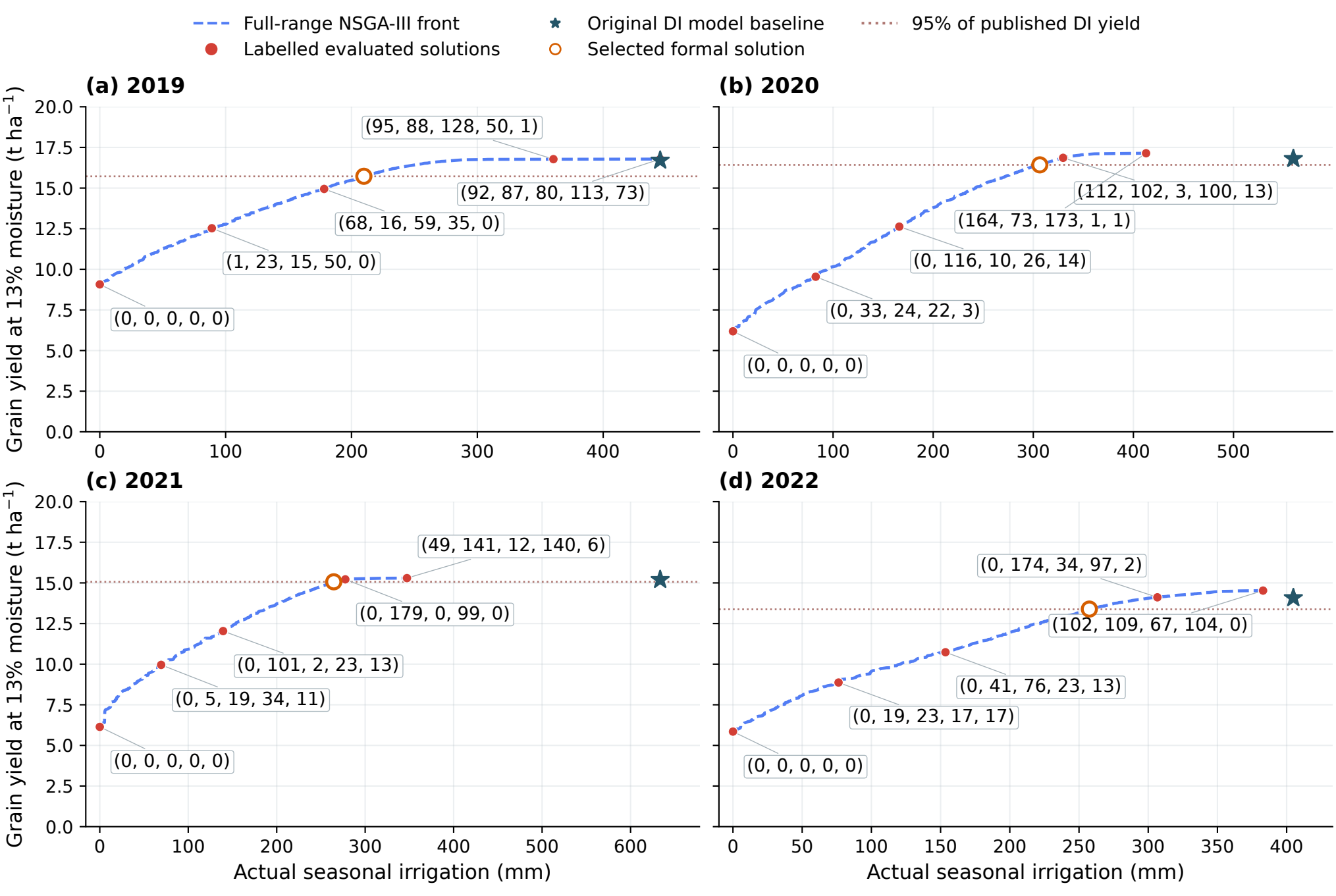

1:1 line
Original calibration target
Thesis-bar rereading
(a) 2019
R^2_{cal} = 0.9926
RMSE = 1.0498 t ha^{-1}
n = 9
(b) 2020
R^2_{cal} = 0.9766
RMSE = 1.1393 t ha^{-1}
n = 10
(c) 2021
R^2_{cal} = 0.9685
RMSE = 1.2520 t ha^{-1}
n = 8
(d) 2022
R^2_{cal} = 0.9966
RMSE = 0.5562 t ha^{-1}
n = 10
Simulated biomass (t ha^{-1})
Digitized DI biomass (t ha^{-1})
0
10
20
30
40
5
15
25
35

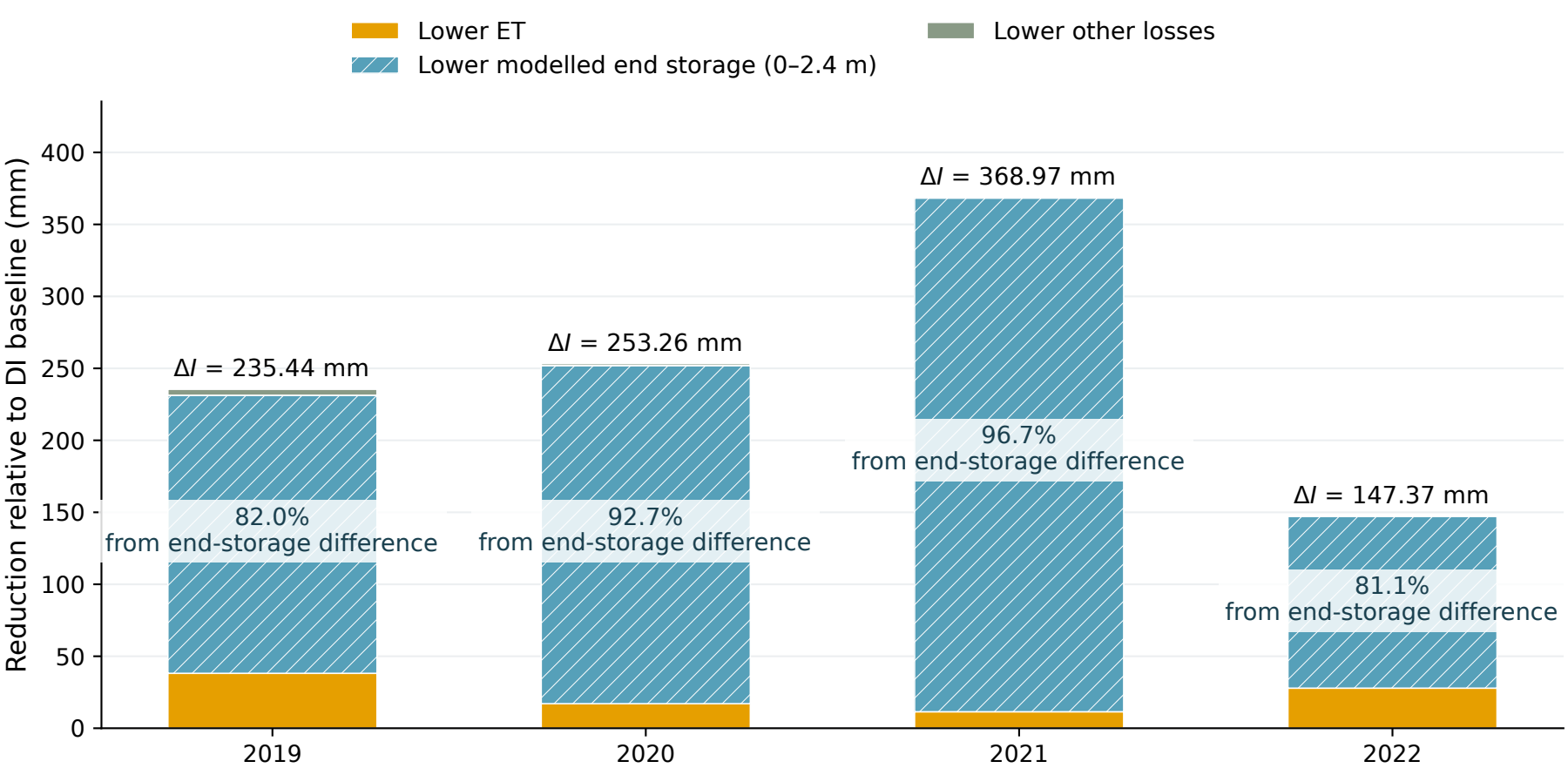

2019
2020
2021
2022
OLS fit
1:1 line
(a) All four years
y = 0.9979x + 0.2412
OLS R² = 0.9892
Pearson r = 0.9946
RMSE = 1.0205 t ha⁻¹
n = 37
(b) Excluding 2022: sensitivity check
y = 1.0070x + 0.2784
OLS R² = 0.9872
Pearson r = 0.9936
RMSE = 1.1457 t ha⁻¹
n = 27
Simulated biomass (t ha⁻¹)
Digitized DI biomass (t ha⁻¹)
Digitized DI biomass (t ha⁻¹)
0
5
10
15
20
25
30
35
40

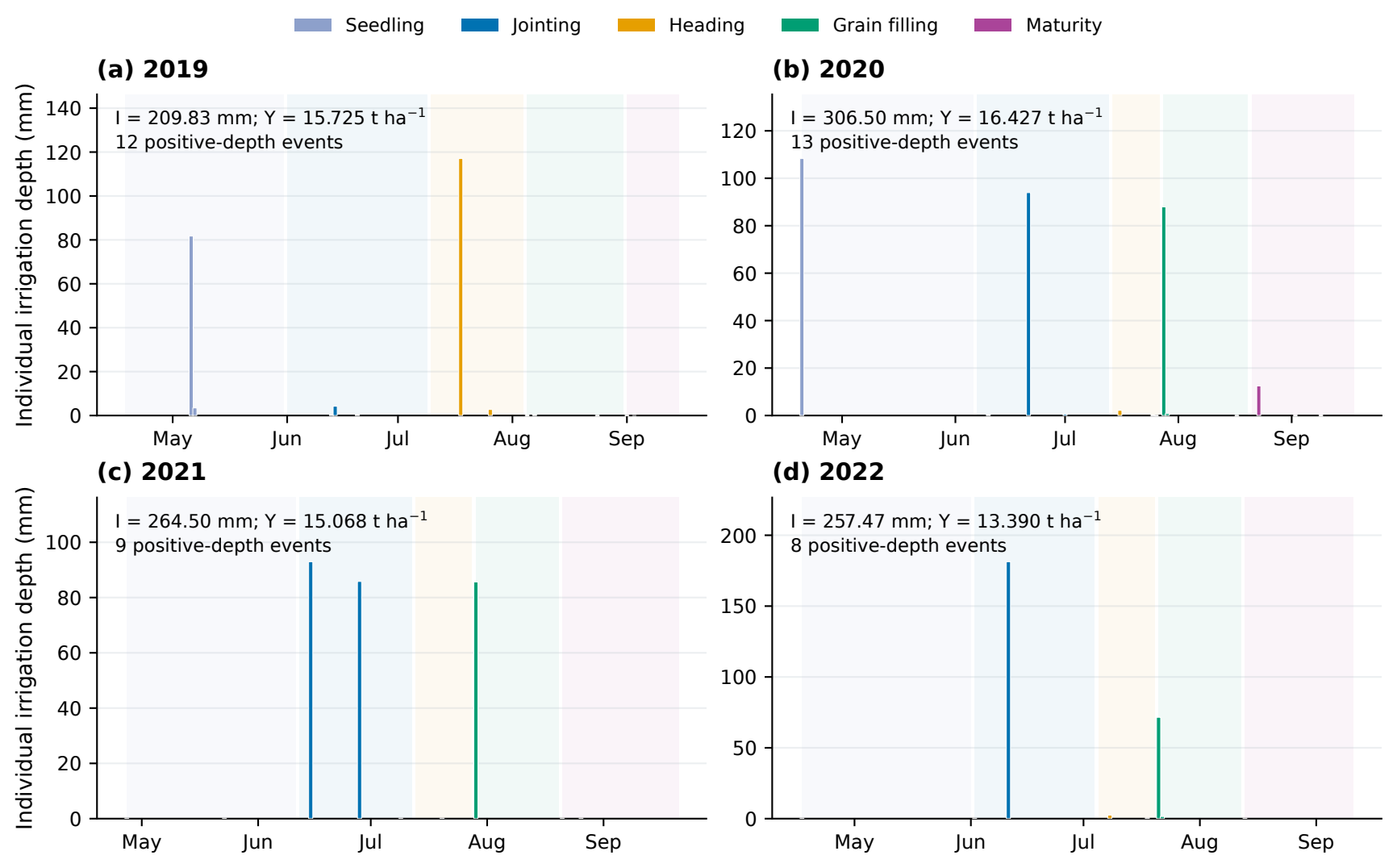

Inflow: irrigation + precipitation
Outflow: ET + water losses
Storage change: final − initial
Seasonal water balance (mm)
800
600
400
200
0
2019
2020
2021
2022

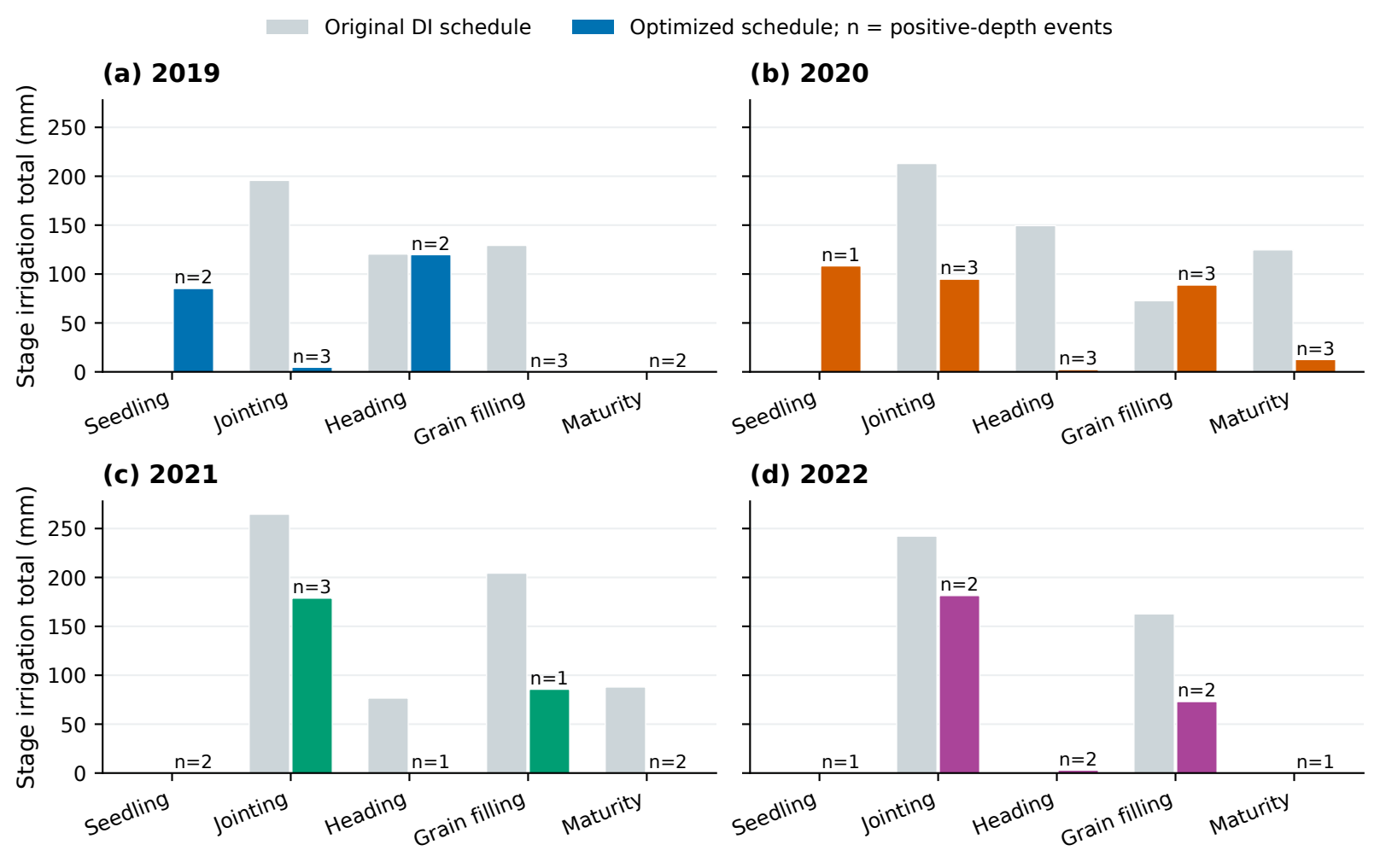

(a) Grain yield
2019 0.060 0.029 0.007 0.013 0.148 0.167 0.393 0.207
2020 0.060 0.035 0.009 0.015 0.147 0.198 0.353 0.208
2021 0.057 0.046 0.003 0.006 0.145 0.150 0.413 0.204
2022 0.052 0.051 0.004 0.008 0.139 0.139 0.426 0.203
(b) Aboveground biomass
2019 0.074 0.035 0.007 0.014 0.185 0.211 0.501 0.000
2020 0.074 0.043 0.007 0.013 0.185 0.253 0.453 0.000
2021 0.071 0.056 0.004 0.008 0.180 0.188 0.521 0.000
2022 0.065 0.061 0.005 0.011 0.171 0.175 0.540 0.000
(c) Seasonal evapotranspiration
2019 0.051 0.063 0.007 0.017 0.882 0.000 0.000 0.000
2020 0.057 0.083 0.008 0.017 0.856 0.000 0.000 0.000
2021 0.050 0.113 0.003 0.007 0.848 0.000 0.000 0.000
2022 0.077 0.149 0.012 0.024 0.762 0.000 0.000 0.000
(d) Maximum canopy cover
2019 0.957 0.048 0.000 0.000 0.000 0.000 0.000 0.000
2020 0.950 0.055 0.000 0.000 0.000 0.000 0.000 0.000
2021 0.900 0.109 0.000 0.000 0.000 0.000 0.000 0.000
2022 0.836 0.177 0.000 0.000 0.002 0.000 0.000 0.000
CCx
CGC
CDC
Senescence
Kcb
WP early
WP late
HI0
Total-effect Sobol index (S_T)
1.0
0.8
0.6
0.4
0.2
0.0

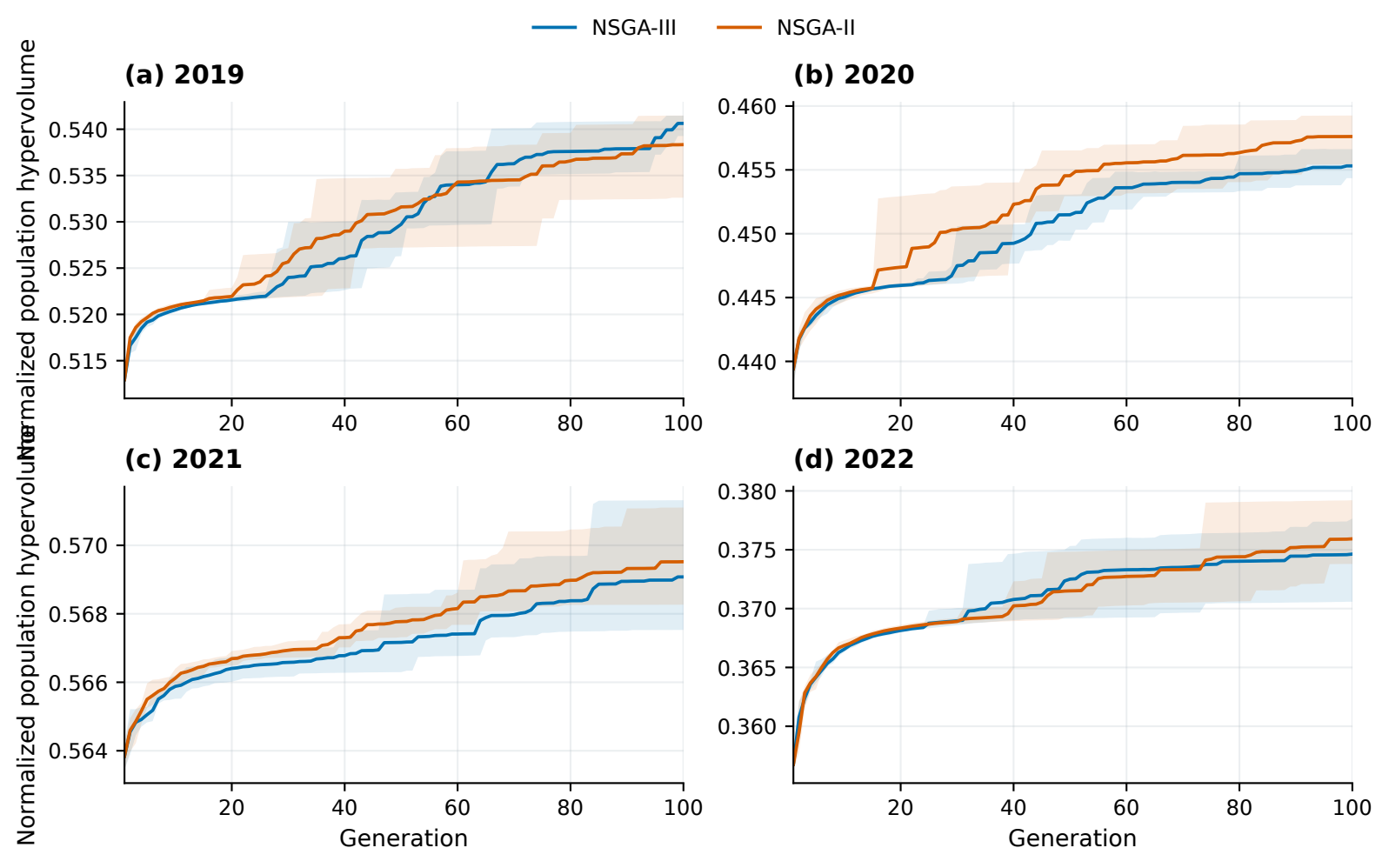

2019
2020
2021
2022
Four-year descriptive mean
(a) First-order effect (S_1)
(b) Total effect (S_T)
WP late
HI0
WP early
Kcb
CCx
CGC
Senescence
CDC
0.0
0.1
0.2
0.3
0.4
0.5
Sobol index
2022: WP early minus Kcb
ΔS_T = 0.000077
Paired 95% interval:
[-0.0139, 0.0138]


Full-range NSGA-III front
Labelled evaluated solutions
Original DI model baseline
Selected formal solution
95% of published DI yield
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Grain yield at 13% moisture (t ha^{-1})
Actual seasonal irrigation (mm)
(0, 0, 0, 0, 0)
(1, 23, 15, 50, 0)
(68, 16, 59, 35, 0)
(92, 87, 80, 113, 73)
(95, 88, 128, 50, 1)
(0, 33, 24, 22, 3)
(0, 116, 10, 26, 14)
(164, 73, 173, 1, 1)
(112, 102, 3, 100, 13)
(0, 5, 19, 34, 11)
(0, 101, 2, 23, 13)
(0, 179, 0, 99, 0)
(49, 141, 12, 140, 6)
(0, 19, 23, 17, 17)
(0, 41, 76, 23, 13)
(102, 109, 67, 104, 0)
(0, 174, 34, 97, 2)


Lower ET
Lower modelled end storage (0–2.4 m)
Lower other losses
Reduction relative to DI baseline (mm)
0
50
100
150
200
250
300
350
400
ΔI = 235.44 mm
82.0%
from end-storage difference
ΔI = 253.26 mm
92.7%
from end-storage difference
ΔI = 368.97 mm
96.7%
from end-storage difference
ΔI = 147.37 mm
81.1%
from end-storage difference
2019
2020
2021
2022


Seedling
Jointing
Heading
Grain filling
Maturity
(a) 2019
I = 209.83 mm; Y = 15.725 t ha^{-1}
12 positive-depth events
(b) 2020
I = 306.50 mm; Y = 16.427 t ha^{-1}
13 positive-depth events
(c) 2021
I = 264.50 mm; Y = 15.068 t ha^{-1}
9 positive-depth events
(d) 2022
I = 257.47 mm; Y = 13.390 t ha^{-1}
8 positive-depth events
Individual irrigation depth (mm)
May
Jun
Jul
Aug
Sep


Original DI schedule
Optimized schedule; n = positive-depth events
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Stage irrigation total (mm)
0
50
100
150
200
250
n=2
n=3
n=2
n=3
n=2
n=1
n=3
n=3
n=3
n=2
n=1
n=3
n=1
n=2
n=1
n=2
n=2
n=1
n=2
n=1
Seedling
Jointing
Heading
Grain filling
Maturity


NSGA-III
NSGA-II
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Normalized population hypervolume
Normalized population hypervolume
Generation
Generation
0.540
0.535
0.530
0.525
0.520
0.515
0.460
0.455
0.450
0.445
0.440
0.570
0.568
0.566
0.564
0.380
0.375
0.370
0.365
0.360
20
40
60
80
100